Journal Style Profile, Evidence Matrix, and Compliance Audit

EAAI Original Research — v3.2.14 evidence lock

1. Journal Style Profile

Rule	Level	Source/evidence	Confidence	Applied
Single-column Word submission	H	Official EAAI guide	High	Yes
Double-anonymised manuscript and separate title page	H	Official EAAI guide	High	Yes
Abstract ≤250 words; 1–6 keywords	H	Official EAAI guide	High	Yes
Highlights: 3–5 bullets, each ≤85 characters	H	Official EAAI guide	High	Yes
Explicit AI contribution and engineering application contribution	H	EAAI aims/scope and desk-reject rules	High	Yes
Methods and Results separated; independent Conclusions	S	Recent Original Research exemplars	High	Yes
Problem formulation before learning architecture	S	Routing/navigation exemplar pool	High	Yes
Results ordered from primary quality to safety, time, training, transfer, robustness, ablation	S	Stable evidence-first convention adapted to study logic	Medium	Yes
Graphical abstract	O	Encouraged but not required	High	Not supplied
Post-hoc composite in main claim	O	Study-specific	Low	No; supplement only

Macrostructure: Introduction → Related work and gap → Problem formulation → Method → Experimental/statistical protocol → Results → Discussion and limitations → Conclusions. Results paragraphs follow purpose → figure → observation → quantitative evidence → statistics → bounded interpretation. Discussion is organised by findings and claim boundaries rather than citation count.

Language profile: precise active/passive mixture; past tense for completed experiments; present tense for figure content and established concepts; restrained hedging; no unverified first/novel/unprecedented claim; no universal PPO superiority. Sample papers influenced only section functions, information order, evidence density, and caption roles.

2. Fact Map

Fact ID	Statement	Source	Type	Status
F01	The frozen protocol identity is multimap_v3_2_14 (v3.2.14).	HANDOFF.md; final_audit_status.json	metadata	verified
F02	The final result file contains 21,648 records and excludes ppo_mlp.	final_audit_status.json; final_results.jsonl	result	verified
F03	Training used 72 procedural mountain maps, 648 tasks, five seeds, and 3,000 episodes for each of seven learning variants.	HANDOFF.md; training summaries	method	verified

Fact ID	Statement	Source	Type	Status
F04	Formal evaluation used 24 unseen procedural maps (216 tasks) and eight independent Copernicus DSM regions (144 tasks).	evaluation matrix; HANDOFF.md	method	verified
F05	Node counts 16, 20, and 24 all occurred in training; evaluation does not test out-of-range scale generalisation.	protocol; HANDOFF.md	limitation	verified
F06	The full actor–critic contains 341,505 parameters for the audited seed-42 checkpoint.	checkpoint audit	method	verified
F07	The traditional comparator is a fixed-slot FlatMLPActorCritic without attention.	implementation; HANDOFF.md	method	verified
F08	Safe priority-weighted coverage assigns zero to unsafe or hard-violation routes.	protocol and analysis code	method	verified
F09	Maps are the independent statistical unit; task, seed, route, and road repetitions are not treated as independent maps.	statistics protocol	statistic	verified
F10	PPO+Pointer mean safe weighted coverage is 0.4858 on unseen synthetic maps and 0.5024 on DSM simulations.	descriptive_metrics.csv	result	verified
F11	PPO+Pointer and A2C+Pointer do not differ significantly on the confirmatory safe coverage outcome.	confirmatory_pairwise.csv	result	verified
F12	ACO, SA, and MILP can exceed PPO+Pointer on safe weighted coverage, with higher online computation.	descriptive_metrics.csv; M05/S02 Source Data	result	verified
F13	Removing the return reserve produces a large, significant coverage loss; the other three individual ablations do not.	confirmatory_pairwise.csv	result	verified
F14	Geographic DSM results are zero-shot simulations and are not real-flight validation or safety certification.	protocol; HANDOFF.md	limitation	verified
F15	The post-hoc 100-point operational score is descriptive and not a confirmatory primary outcome.	operational analysis manifest	limitation	verified

3. Claim Map and Evidence Matrix

Claim	Maximum statement	Evidence	Evidence type	Max strength	Boundary

Claim	Maximum statement	Evidence	Evidence type	Max strength	Boundary
C01	The method enforces return-aware resource feasibility before action selection.	F08; Fig. 1; implementation	method fact	show	No formal safety certification
C02	Pointer attention supports variable-length candidate selection within the trained node-count range.	F05; Fig. 1	method fact	supports	No claim beyond 16/20/24 nodes
C03	PPO+Pointer strongly improves safe weighted coverage over Flat-MLP PPO.	F10; Fig. 2; pairwise statistics	result	shows	Simulation domains only
C04	PPO+Pointer and A2C+Pointer attain statistically similar confirmatory safe coverage.	F11; Fig. 2	result	indicates	Non-significance is not proof of equivalence
C05	PPO+Pointer has a stronger combined training profile than A2C+Pointer.	Fig. 5; D6/D7 frozen scores	result	indicates	Mechanism not causally isolated
C06	Return reserve is the dominant independently tested safety-related component.	F13; Fig. 8	result	supports	Other submasks were not separately ablated
C07	Classical solvers expose a quality–time trade-off rather than universal PPO dominance.	F12; Fig. 4	result	shows	Hardware-specific time values
C08	The policy transfers zero-shot to independent geographic DSM simulations.	F04; F14; Fig. 6	result	shows	Not physical-domain transfer
C09	Hidden mismatch degrades performance but preserves a PPO advantage over Flat-MLP PPO.	Fig. 7; robustness tables	result	indicates	Finite perturbation catalogue
C10	The study provides reproducible simulation evidence for engineering planning decisions.	F01–F15; package manifest	synthesis	supports	Requires independent external and flight validation

4. Figure Storyline

Figure	Question	Maximum claim	Cannot prove
Fig. 1	How are variable candidates and return constraints coupled?	Feasibility is filtered before action selection.	Formal or physical safety
Fig. 2	Does the method improve safe coverage?	Large gain over Flat-MLP PPO; similarity to A2C.	Universal coverage dominance
Fig. 3	Are routes safe and resource-aware?	Primary simulated feasibility and bounded utilisation.	Certified aircraft margins
Fig. 4	What is the online quality–time trade-off?	Neural methods are much faster than ACO/MILP here.	Hardware-independent latency

Figure	Question	Maximum claim	Cannot prove
Fig. 5	How do learning trajectories differ?	PPO has stronger frozen D6/D7 profile.	Causal PPO mechanism
Fig. 6	Does the policy transfer across geographic DSM contexts?	Zero-shot simulation transfer.	Sim-to-real or flight transfer
Fig. 7	How does performance change under shifts?	Finite known/hidden perturbation robustness.	Open-world robustness
Fig. 8	Which components have independent evidence?	Return reserve is dominant.	Independent submask effects

5. Final Compliance Audit

Audit area	Evidence	Status
Scientific	final_results.jsonl = 21,648; passed=true; ppo_mlp_absent=true	PASS
Statistics	Map is independent unit; paired tests and outer-map bootstrap	PASS
Abstract	206 words	PASS
Keywords	5	PASS
Highlights	5 bullets; max 75 characters	PASS
Anonymity	Author identities omitted from manuscript; separate title page	PASS WITH PLACEHOLDERS
References	DOI/official metadata register and claim map supplied	PASS; access-limited full texts marked
Generative images	None used; plots redrawn from frozen Source Data	PASS
Data/code	Anonymous/permanent repository links missing	AUTHOR INPUT REQUIRED
Funding/CRediT/COI	No author-supplied facts	AUTHOR INPUT REQUIRED
Page limit	To be verified after rendered layout	PENDING RENDER AUDIT
Current journal rule	Recheck author guide and AI disclosure at submission	AUTHOR CHECK

6. Author Verification Queue

- [AUTHOR CHECK 01] Author and affiliation identities.
- [AUTHOR CHECK 02] Hardware and software versions.
- [AUTHOR CHECK 03] Anonymous and permanent repository links/licences.
- [AUTHOR CHECK 04] Copernicus derivative redistribution terms.
- [AUTHOR CHECK 05] Funding, CRediT, competing interests, acknowledgements.
- [AUTHOR CHECK 06] Current EAAI AI disclosure wording at submission.